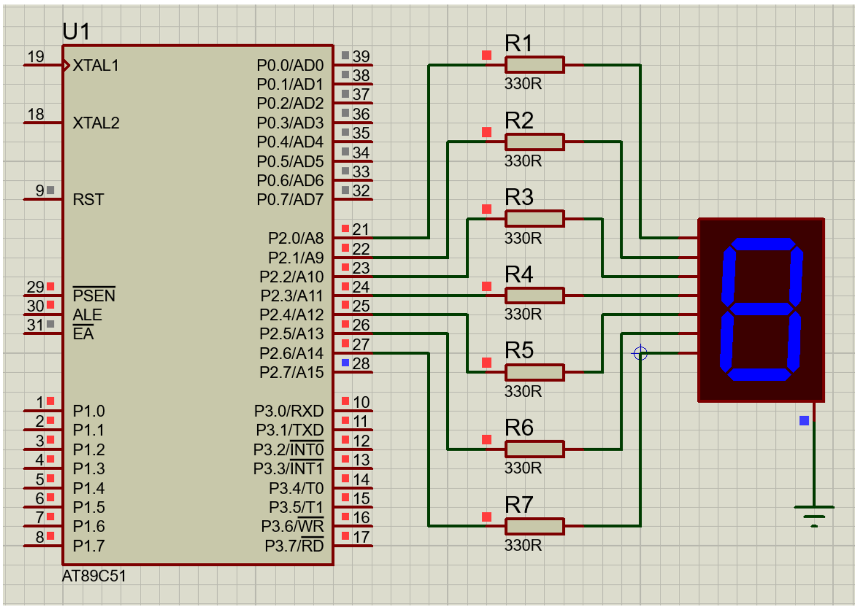
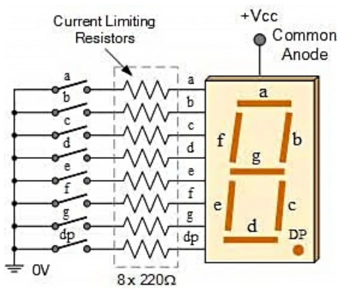
3️⃣ 7-Segment Display - Print 0 to 9

Aim: To write & execute a program to interface a 7-segment display with the 8051 microcontroller & display digits (0-9).





Apparatus Required:

- 8051 microcontroller
- 7-segment display (Common Cathode)
- Resistors (330Ω)
- Breadboard
- Connecting wires
- Keil uVision IDE

Working Principle:

- Each segment is controlled individually using the microcontroller's port pins.
- By sending proper binary patterns, digits can be displayed.
- Logic 1 → Segment ON (glows).
- Logic 0 → Segment OFF.
- 7-segment displays consist of 7 LEDs labeled a, b, c, d, e, f, and g, used to display digits (0-9).
- Ex: To display the digit 0, segments a, b, c, d, e, f are ON. The machine sends binary data to control these segments.

Operations:

- Each segment is connected to the O/P port pins of the 8051 microcontroller.

- Each pin controls one segment of the display.
- By sending specific binary patterns to the port, different combinations of segments are turned ON/OFF, forming digits (0-9).

Common Cathode Working:

- All cathodes are connected to GND.
- A segment glows when logic High (1) is applied.
- Eg: to display 0 → segments a, b, c, d, e, f are ON & g is OFF.
- Binary pattern: 00111111 → Hex: 0x3F.
- Common Cathode forward biased → LED glows.

Truth Table (Common Cathode Hex Codes):

Digit	**g**	**f**	**e**	**d**	**c**	**b**	**a**	**Binar**
0	0	1	1	1	1	1	1	00111
1	0	0	0	0	1	1	0	0000
2	0	1	0	1	1	0	1	01011
3	0	1	0	0	1	1	1	01001
4	0	1	1	0	0	1	1	01100
5	0	1	1	0	1	0	1	01101
6	0	1	1	1	1	0	1	011111
7	0	0	0	0	1	1	1	0000
8	0	1	1	1	1	1	1	011111
9	0	1	1	0	0	1	1	01100

Program Code:

```c
#include <reg51.h>

// Standard hardware delay
void delay(unsigned int ms)
{
    unsigned int i, j;
    for(i = 0; i < ms; i++)
        for(j = 0; j < 1275; j++);
}

void main()
{
    int i;

    // Use 'unsigned char' for 8-bit port data.
    // Adding the 'code' keyword tells Keil to store this array in ROM (Flash)
    // instead of wasting the 8051's extremely tiny 128-byte RAM.
    unsigned char code num[] = {0x3F, 0x06, 0x5B, 0x4F, 0x66,
                                0x6D, 0x7D, 0x07, 0x7F, 0x67};

    // The infinite loop traps the microcontroller so it counts continuously
    while(1)
    {
        for(i = 0; i < 10; i++)
        {
            P2 = num[i];  // Output the hex value to the display
            delay(100);   // Wait so the human eye can read it
        }
    }
}
```

Result: The 7-segment display was successfully interfaced with the 8051 microcontroller & digits were displayed correctly.

Precautions:

- Always use the correct type (common anode/cathode).
- Always use current-limiting resistors.
- Ensure proper pin connections.
- Avoid incorrect voltage supply.